

RESEARCH ARTICLE

An Archive-Based Multi-Objective Arithmetic Optimization Algorithm for Solving Industrial Engineering Problems

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ABSTRACT This research proposes an Archive-based Multi-Objective Arithmetic Optimization Algorithm (MAOA) as an alternative to the recently established Arithmetic Optimization Algorithm (AOA) for multi-objective problems (MAOA). The original AOA approach was based on the distribution behavior of vital mathematical arithmetic operators, such as multiplication, division, subtraction, and addition. The idea of the archive is introduced in MAOA, and it may be used to find non-dominated Pareto optimum solutions. The proposed method is tested on seven benchmark functions, ten CEC-2020 mathematic functions, and eight restricted engineering design challenges to determine its suitability for solving real-world engineering difficulties. The experimental findings are compared to five multi-objective optimization methods (Multi-Objective Particle Swarm Optimization (MOPSO), Multi-Objective Slap Swarm Algorithm (MSSA), Multi-Objective Ant Lion Optimizer (MOALO), Multi-Objective Genetic Algorithm (NSGA2) and MultiObjective Grey Wolf Optimizer (MOGWO) reported in the literature using multiple performance measures. The empirical results show that the proposed MAOA outperforms existing state-of-the-art multi-objective approaches and has a high convergence rate.

INDEX TERMS Arithmetic optimization algorithm (AOA), archive-based multi-objective arithmetic optimization algorithm (MAOA), multi-objective problems, engineering optimization.

I. INTRODUCTION

Metaheuristics are efficient in solving complex real-world problems mainly due to their black-box optimization nature and gradient-free mechanism. Recently, various metaheuristic methods have been proposed: based on natural processes, collective behavior, or scientific rules, such as Differential Evolution (DE) [1], Ant Colony Optimization (ACO) [2],

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Particle Swarm Optimization (PSO) [3], Reptile Search Algorithm (RSA) [4], Dynamic Arithmetic Optimization Algorithm (DAOA) [5], Transient Search Optimization (TSO) [6], Dynamic Water Strider Algorithm (DWSA) [7], Aquila Optimizer (AO) [8], Stochastic Paint Optimizer (SPO) [9] and Group Teaching

Optimization (GTO) [10] algorithm. Furthermore, several researchers used these algorithms to solve various problems [11], [12], [13], [14], [15], [16], [17]. These studies show that metaheuristic algorithms can solve problems with high precision and a reasonable amount of time when used to solve complex optimization problems. Moreover, metaheuristic algorithms have many advantages over analytical methods, including ease of implementation, a straightforward structure, good accuracy, and an appropriate execution time.

Uncertainties in weights, geometry, structural properties, manufacturing processes, and operating conditions, among other things, are still present in engineering problems [18]. A deterministic optimization can yield a feasible solution for problems with poor uncertainties. On the other hand, deterministic optimization may result in inefficient or inconsistent solutions for issues with high uncertainties, raising the probability of design failure. Several nondeterministic optimization algorithms have been designed to address this matter and successfully solve various design problems in real-world engineering [19], [20], [21].

Since the advent of computing, the field of computeraided design has arisen, allowing engineers to use computers to solve engineering problems. Designers will use a computer to model the dilemma and its variables in the early stages. As a result, there was no need to create a prototype and test it on real-world problems. Computer software, for example, can make the 3D outline of a car's body when simulating wind drag [22]. In this scenario, the programmers were more concerned with architecture than simulation. This resulted in a substantial reduction in the overall expense of the design method and human interaction.

The vast majority of the real-world optimization problems are multi-objective by nature [23]. Multi-objective optimization is a branch of decision-making that balances various frequently disputed goals [24]. It has been used in various research fields, including physics, economics, and logistics, where optimum choices must be considered in balancing between two or more competing goals. A single-objective approach will only represent one part of a multi-objective problem if used to solve it. A single-objective algorithm, for example, cannot wholly describe the hydrological mechanism and function in optimizing hydrological design variables [25].

The task of solving one objective function problem is easy. The optimal solution is the product of such problems, and the optimization methods are assessed on the nature of the current solutions

using the specified objective functions [26]. However, many objective functions compete with one another in multi-objective problems. There is no one-size-fits-all approach that is optimal for all of them simultaneously. Therefore, the decision-makers are searching for the established alternative instead of the best solution in these situations. The Pareto solutions are a subset of these types of solutions. As a result, in multi-objective problems, optimality is renewed by the method of Pareto optimality. The Pareto optimal (besides known as effective, non-dominated, or noninferior) solutions cannot increase their efficiency in one objective function without affecting the achievement of at least one of the others; this is the strategy of dealing with more

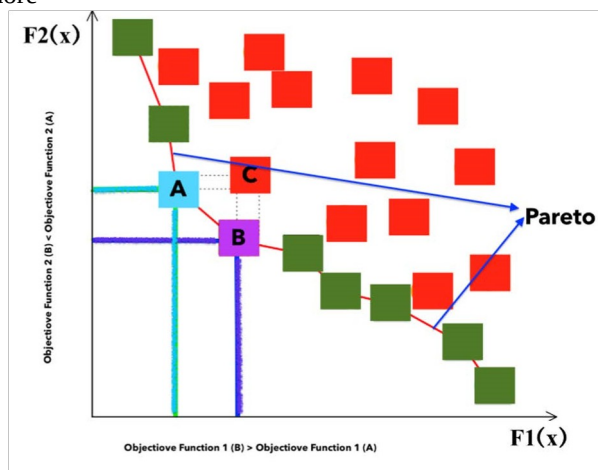


FIGURE 1. Pareto dominance.

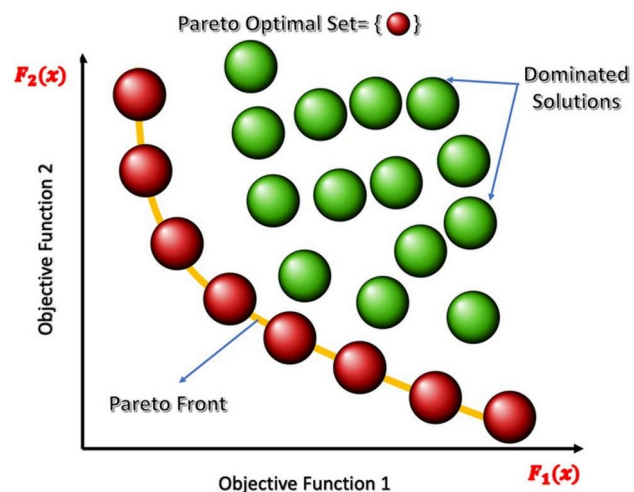


FIGURE 2. Pareto optimal solutions.

than objective function. The Pareto set is the collection of Pareto optimal solutions [27].

The following is an example of a general multi-objective optimization problem:

$$\text{minimize } F(x) = (f_1(x), f_2(x), f_i(x), \dots, f_t(x))^T, x \in R^k \quad (1)$$

where x is the possible decision values, $x \in R^k$ is the decision variable, k is the number of decisions, $F: R^k \rightarrow R^m$ is a scalar objective function with multidimensional objective components $f_i(x)$, $i =$

$1, 2, \dots, t \geq 2$, and R^m is the objective range. Thus, $F(x)$ contains t objectives, and the given problem includes k decisions. The decision range $\subset R^k$ is a linked and closed area. Moreover, all objective functions are continuous throughout the decision space; thus, a multi-objective function in any real optimization problem is a continuous multivariable problem [28].

Most recent algorithms have rules in place to solve problems with multiple goals. However, the famous

no one best method or algorithm can solve all optimization. However, new algorithms are best suited to unconstrained problems. This means existing algorithms can be improved problems and cannot handle various constraints without Pseudo-Code of the MAOA Algorithm

Initialize the MAOA parameters

Initialize the soLutions' positions

Evaluate the fitness function for given soLutions

Obtain the non-dominated answers and create the archive

While (Iteration < Max-Iteration)

 Update MOA and MOP with Eqs (10) and (11)

For each soLution

Find

 Lea

 der

for

 eac

 h

 posi

 tion

 Create a random number for r_1 , r_2 ,
 r_3 between $[0,1]$ **if** $r_1 > MOA$ then **the**
exploration step will be active

If $r_2 > 0.5$

 Update the position with the first rule of Eq (12)

Else

 Update the position with the second
rule of Eq (12) **End if**

Else if $r_3 > MOA$ then **the exploitation**

Step will be active

If $r_2 > 0.5$

 Update the position with the first rule of Eq (13)

Else

 Update the position with second
rule of Eq (13) **End if**

End if

End for

End for

Evaluate the objective values for aLL popuLation

Obtain the non-dominated answers

Update the archive according to the found non-dominated answers

If the archive is compLeted

 Use the grid mechanism to deLete the current archives

 Add the new answer to the archive

Endif

If any of the newLy added answers to the archive is pLaced outside of the hypercubes

 Update the grids

Return

En

Archive

d

if

End

using specific components. The multi-objective

while;

variant of the recently introduced Arithmetic

Optimization Algorithm (AOA) is proposed in this paper to solve both unconstrained and constrained problems. The mathematical operators in math sciences became the main inspiration for the conventional AOA. The Archive-based Multi-objective Arithmetic Optimization Algorithm (MAOA) uses an archive and leader selection process used in Multi-objective Particle Swarm Optimization. The experiments are conducted on several various real-world engineering problems. The proposed MAOA proved its ability to solve multiple complex problems compared to other state-of-the-art multiobjective methods. The remainder of the article is laid out as follows:

Most recent algorithms have rules in place to solve problems with multiple goals. However, the well-known No-Free-Lunch (NFL) theorem proves that none of such algorithms can solve all optimization problems. This means existing algorithms can be improved or new ones proposed to help solve particular problems. New algorithms are best suited to unconstrained problems and cannot handle various kinds of constraints without the use of specific components. The multi-objective variant of the recently introduced Arithmetic Optimization Algorithm (AOA) is proposed in this paper to solve both unconstrained

and constrained problems. The mathematical operators in math sciences became the main inspiration for the conventional AOA. The Archivebased Multi-objective Arithmetic Optimization Algorithm (MAOA) uses an archive and leader selection process used in Multi-objective Particle Swarm Optimization. The experiments are conducted on several various real-world engineering problems. The proposed MAOA proved its ability to solve multiple complex problems compared to other well-known multi-objective methods. The remainder of the article is laid out as follows:

Section 2 presents the multi-objective optimization (MOO) theory. Section 3 introduces Arithmetic Optimization Algorithm and its multi-objective version. Performance metrics and results of test functions and engineering problems are discussed in Section 4. Finally, this study is concluded in the final Section.

II. LITERATURE REVIEW

The Multi-objective Optimization (MOO) theory and current meta-heuristic techniques are discussed in this Section.

Due to the unary goal and the existence of just one most satisfactory solution in single-objective

TABLE 3. The statistical results of mathematical functions for GD performance metric.

Functions		Algorithm			
		MOPSO	MSSA	MOALO	MAOA
ZDT1	Ave	1.9932E-03	1.2035E-02	1.9644E-03	1.8467E-03
	SD	5.0496E-03	1.9168E-03	1.7358E-04	2.5549E-03
	WRT	+	+	-	
ZDT2	Ave	1.4749E-01	1.1652E-02	4.0621E-04	2.1622E-04
	SD	6.5211E-02	2.1916E-03	1.9757E-05	1.6201E-05
	WRT	+	+	+	
ZDT3	Ave	2.0489E-04	1.6015E-02	1.7253E-03	2.1406E-03
	SD	2.7322E-05	2.0859E-03	1.1636E-03	3.4858E-03
	WRT	-	+	+	
ZDT4	Ave	4.1299E+00	6.0099E-01	2.0045E+00	2.6285E-04
	SD	3.9519E+00	3.1030E-01	9.6098E-01	3.2461E-05
	WRT	+	+	+	
ZDT6	Ave	2.5805E-02	2.1047E-02	3.3034E-02	1.6268E-03
	SD	5.8008E-02	9.6106E-03	2.0337E-02	5.0708E-03
	WRT	+	+	+	
DTLZ2	Ave	6.8600E-03	9.2866E-03	2.1134E-02	1.8112E-01
	SD	9.0878E-04	2.4582E-03	1.0861E-02	1.0018E-02
	WRT	-	+	+	
DTLZ4	Ave	1.0097E-02	1.3120E-02	3.6328E-02	2.1398E-01
	SD	2.4570E-03	3.8723E-03	1.9865E-02	2.3231E-02
	WRT	-	+	+	
W ⁺ /W ⁻		8/6	14/0	14/1	
+/-/=		4/3/0	7/0/0	7/0/0	

problems, the global optimum in single-objective problems is only one solution. When only one target is in consideration, comparing solutions with relational operators is easy. Because of the characteristics of such problems, evaluating potential solutions and selecting the best one is easy for optimization problems. Multi-objective evaluation is a branch of multiple approaches to solving those deal with mathematical programming issues that require the evaluation of numerous objective functions at the same time.

As a result, research into effective and reliable multiobjective optimization algorithms is needed. Many interesting experiments have been conducted in multi-objective optimization to deal with complicated multi-objective nonlinear problems. A novel congestion control method is proposed for Wireless Sensor Networks in [30]. Congestion management is conducted by determining the optimum rate by a simple Poisson method. The current techniques used to solve this problem have high complications and power consumption due to retransmission. To address these problems, the multi-objective optimization technique (PSOGSA) is used to suggest a congestion control strategy [30]. For speed optimization and controlling the data arrival rate from each child node to the parent node. The proposed method used two main search

mechanisms; Particle Swarm Optimization (PSO) and Gravitational Search Algorithm (GSA).

A multi-objective optimization function considers the energy of the connection in its fitness value. Priority-dependent transmission is allowed because the optimization strategy controls the arrival rate based on priority: available performance bandwidth and child node capacity. Adjusting the speed to the desired value is used to alleviate congestion. The results of simulations show that the proposed method outperforms current methods.

Due to its considerable impact on grid security and power dispatching, researchers have recently paid growing awareness to reliable and consistent wind speed estimation. On the other hand, most current research has concentrated solely on improving precision or stability, with only a few studies covering both problems simultaneously. This job is difficult due to the intermittency and dynamic variations of wind speed. This paper proposes an innovative hybrid method based on the Multi-Objective Whale Optimization Algorithm [31], called MOWOA, consisting of 4 modules: optimization, data preprocessing, forecasting, and assessment. The suggested MOWOA's optimization module is used to configure the weights and thresholds of the neural network utilized in the predicted module to achieve high

TABLE 4. The statistical results of mathematical functions for IGD performance metric.

		<i>MOPSO</i>	<i>MSSA</i>	<i>MOALO</i>	<i>MAOA</i>
ZDT1	Ave	8.0879E-04	4.5908E-03	1.5144E-02	5.0981E-04
	SD	1.4764E-03	1.1201E-03	2.1178E-03	6.5912E-05
	WRT	+	+	+	
ZDT2	Ave	5.2023E-02	5.2059E-03	2.1623E-02	5.0588E-04
	SD	9.5007E-03	1.6220E-03	1.7638E-03	4.2200E-05
	WRT	+	+	+	
ZDT3	Ave	2.6241E-04	5.1290E-03	3.7118E-03	2.5010E-04
	SD	3.9884E-05	8.9560E-04	1.6638E-03	3.4233E-05
	WRT	+	+	+	
ZDT4	Ave	7.0747E-01	1.8230E-01	6.1546E-01	9.0817E-04
	SD	3.1784E-01	9.5457E-02	2.4280E-01	1.7711E-04
	WRT	+	+	+	
ZDT6	Ave	8.2281E-03	2.0670E-03	3.4649E-03	6.2074E-04
	SD	2.4862E-02	8.0527E-04	1.8983E-03	9.2925E-05
	WRT	+	+	+	
DTLZ2	Ave	4.6940E-04	3.0425E-03	3.2518E-03	6.8738E-03
	SD	2.6098E-05	3.7517E-04	4.6848E-04	3.1549E-04
	WRT	-	+	+	
DTLZ4	Ave	1.6840E-03	8.3377E-03	1.1027E-02	2.1954E-02
	SD	9.4057E-05	1.1001E-03	2.9491E-03	1.2775E-03
	WRT	-	+	+	
w^+/w^-		10/4	14/0	14/0	
$+/-/=$		5/2/0	7/0/0	7/0/0	

precision and confidence for wind speed prediction while overcoming the shortcomings of single-objective algorithms. The findings show that the proposed MOWOA outperforms the other sixteen methods.

Cloud computing service arrangements are a complicated optimization problem for a big company, given many users and owned services. Sheikholeslami and Navimipour [32] considered three competing goals: optimizing resources for consumers and suppliers and identifying the best solution at the write time. Since the highly efficient converging to the Pareto front produces a well-distributed collection of nondominated solutions, a Crowding Distance is used in the Multi-Objective Particle Swarm Optimization (MOPSO-CD) to address the problem. Furthermore, the best-obtained solution is often determined using the fuzzy set approach. Again, the suggested method's efficiency is related to two other multi-objective algorithms' performance to demonstrate that it is powerfully efficient against them. Finally, the experiment findings revealed that the system increases the speed at which the services distribution algorithm is executed while also creating significant income for consumers and suppliers and increasing resource efficiency.

A multi-objective method-based energy-aware, called EA-MOA, is proposed in [33] to solve a complex problem (flow shop scheduling) when considering service energy consumption. Simultaneously, two goals are considered: the reduction of the makespan and the reduction of energy demand. First, each solution is expressed by two vectors in the proposed EA-MOA: the priority vector for machine assignment and the scheduling vector. Second, four different decoding techniques are tested to account for these goals. Third, to use the origin solutions from the archive of the Pareto package, two dynamic crossover mechanisms, Single-point Pareto-based crossover (SPBC) and Two-point Pareto-based crossover (TPBC), is performed. After that, eight neighborhood search processes and an adaptive neighborhood selection approach are constructed based on the problem structure. Besides, a right-shifting protocol is used to reduce the running time for all devices, thus increasing the specified solution's energy consumption target. Finally, the proposed EA-MOA is analyzed against several well-known benchmark cases. Finally, the highly successful proposed EA-MOA method is tested and analyzed

TABLE 5. The statistical results of mathematical functions for MS performance metric.

Functions		Algorithm			
		MOPSO	MSSA	MOALO	MAOA
ZDT1	Ave	1.0070E+00	9.0233E-01	3.0049E-01	1.0828E+00
	SD	2.3779E-02	1.0585E-01	5.1645E-02	1.7251E-02
	WRT	+	+	+	
ZDT2	Ave	7.4311E-02	7.7491E-01	3.6133E-02	1.0000E+00
	SD	2.3499E-01	1.0567E-01	4.3001E-02	0.0000E+00
	WRT	+	+	+	
ZDT3	Ave	9.9865E-01	8.7507E-01	7.1037E-01	1.0303E+00
	SD	1.7888E-03	6.5264E-02	1.1021E-01	6.0899E-02
	WRT	-	+	+	
ZDT4	Ave	4.6863E-01	1.4795E-01	2.3408E-01	1.0000E+00
	SD	1.4819E+00	1.7705E-02	4.8105E-02	0.0000E+00
	WRT	+	+	+	
ZDT6	Ave	1.1170E+00	1.1361E+00	1.1199E+00	1.2932E+00
	SD	5.3983E-01	1.8535E-01	5.8815E-01	1.4093E-01
	WRT	+	+	+	
DTLZ2	Ave	1.0976E+00	1.8014E-01	1.5348E-01	2.7164E+00
	SD	3.0869E-02	1.6986E-01	6.5189E-02	7.8217E-02
	WRT	-	+	+	
DTLZ4	Ave	1.2437E+00	2.5352E-01	3.3046E-01	2.7172E+00
	SD	1.4187E-01	2.5166E-01	1.8953E-01	1.0237E-01
	WRT	+	+	+	
W^+/W^-		14/0	14/0	14/0	
VOLUME 10, 2022, 2424-2434		7/0/0	7/0/0	7/0/0	

TABLE 6. The statistical results of mathematical functions for S performance metric.

Functions		Algorithm			
		MOPSO	MSSA	MOALO	MAOA
ZDT1	Ave	1.1319E-02	1.2165E-02	5.1688E-03	1.4712E-02
	SD	1.0762E-03	3.3123E-03	1.9499E-03	5.9223E-03
	WRT	-	+	+	
ZDT2	Ave	7.3295E-02	1.2250E-02	8.6826E-02	6.2168E-02
	SD	2.3178E-03	6.3466E-03	1.9868E-03	1.9380E-03
	WRT	+	+	+	
ZDT3	Ave	1.3270E-02	1.4228E-02	1.4655E-02	1.2602E-02
	SD	2.3561E-03	6.5709E-03	9.2716E-03	2.1021E-03
	WRT	+	+	+	
ZDT4	Ave	4.3804E-03	2.2121E-03	1.4977E-02	2.1446E-02
	SD	1.3852E-02	1.9168E-03	2.5415E-02	1.8980E-03
	WRT	+	+	+	
ZDT6	Ave	1.9109E-02	1.8684E-02	2.4910E-02	1.9105E-02
	SD	3.0071E-02	1.2219E-02	1.2571E-02	1.0440E-02
	WRT	+	+	+	
DTLZ2	Ave	5.7741E-02	9.1604E-03	1.1726E-02	2.0007E-01
	SD	4.9507E-03	5.5147E-03	5.3284E-03	2.7188E-02
	WRT	+	+	-	
DTLZ4	Ave	6.1316E-02	1.5061E-02	1.8314E-02	2.4654E-01
	SD	5.3260E-03	7.5177E-03	1.0509E-02	4.4854E-02
	WRT	-	+	+	
W^+/W^-		14/0	8/6	12/2	
$+/-/=$		7/0/0	4/3/0	6/1/0	

with other qualified algorithms by studying experimental effects.

Mirjalili et al. [34] proposed the Multi-Objective version of Ant Lion Optimizer (MOALO), which is modeled by the hunting behavior of antlions and the way ants interact with them. The non-dominated Pareto optimum answers thus far are stored in a repository first. In this case, a roulette wheel mechanism is used to choose the best solutions from this repository. The roulette wheel chooses the best multi-objective search areas based on how many antlion search solution has. Several standard unconstrained and restricted test functions demonstrate the method's effectiveness under consideration. The approach is also used to solve a number of multi-objective engineering design problems.

A multi-objective version of the Crystal Structure Algorithm (MOCryStAl) was recently presented by Khodadadi et al. [35], and it was motivated by the ideas of crystal structure creation. This algorithm's three primary mechanisms were the grid, leader selection, and archive. The efficacy of the presented method is evaluated using

mathematic problems and real-world engineering design challenges. The findings suggest that the proposed solutions can produce remarkable results when applied to the multi-objective challenges under examination.

Francis Ysidro et al. [36] compared two solutions to multi-objective problems in 1881, and Vilfredo Pareto [37] expanded on this in 1964. The following is multi-objective optimization (MOO) for minimization problems [38]:

$$\text{Maximization: } F(xE) = \{f_1(xE), f_2(xE), \dots, f_z(xE)\} \quad (2)$$

$$\begin{aligned} \text{Subject to: } & g_i(xE) \geq 0 \quad i = 1, 2, \dots, m \quad (3) \quad h_i(xE) = 0 \quad i \\ & = 1, 2, \dots, p \quad (4) \quad L_i \leq x_i \leq U_i \quad i = 1, 2, \dots, n \end{aligned} \quad (5)$$

where, the number of variables, inequality and equality constraints, objective functions are described as m , p , n and z , respectively. The boundaries of i th variable are indicated by L_i and U_i .

A. PARETO DOMINANCE

In this part, there are a couple of solutions (i.e., x_E and E_y) with fitness functions values $x_E = (x_1, x_2, \dots, x_k)$ and $E_y = (y_1, y_2, \dots, y_k)$ that accomplished the given constraints of a multi-objective problem. The main aim here is to determine the smallest objective value, the solution x_E is found to dominate E_y (denote as $x_E < E_y$), if the current values of E_y are not smaller than the current cost value of x_E , and at least one component of x_E must be smaller than that of E_y (see Fig. 1).

This definition may be expressed numerically as follows [39]:

$$\forall i \in \{1, 2, \dots, k\} : \quad f_i(x_E) \leq f_i(E_y) \wedge \exists i \in \{1, 2, \dots, k\} : f_i(x_E) < f_i(E_y)$$

The concept of Pareto optimality, as per this concept of Pareto domination, is as follows:

B. PARETO OPTIMALITY (PO)

A solution $x_E \in X$ is called Pareto-optimal if and only if:

$$\nexists E_y \in X | E_y < x_E$$

The Pareto optimum is a set of non-dominated solutions to complex problems that are presented as follows:

C. PARETO OPTIMAL SET (POS)

The Pareto optimum set (P_s) for a given MOO is described as Eq. (8). This set comprises all viable solutions that cannot be controlled by anyone else plausible option. The collection of Pareto optimum solutions is depicted in Fig. 2.

$$P_s := \{x_E, E_y \in X | \nexists E_y < x_E\}$$

The following is the definition and description of the Pareto optimal front:

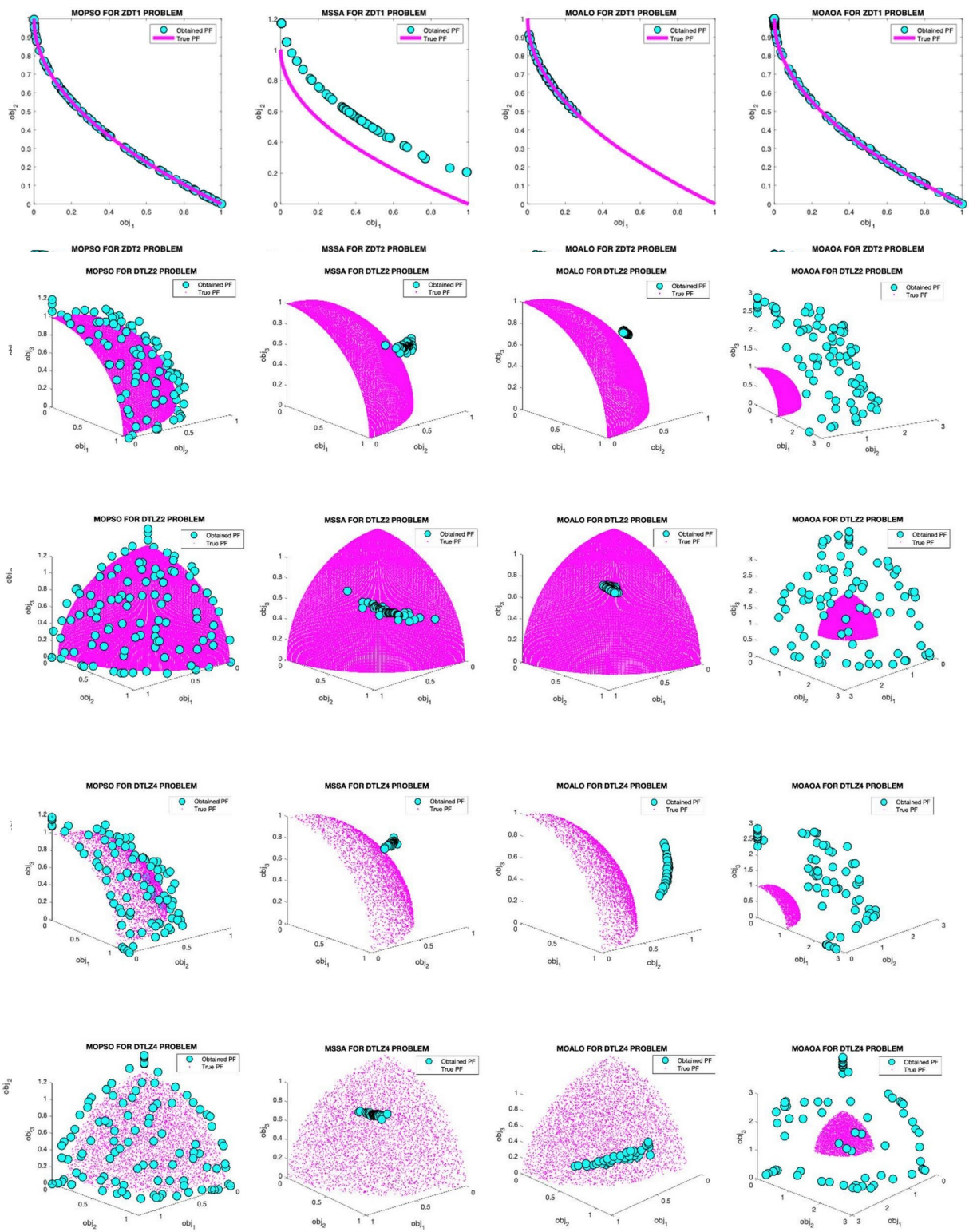


FIGURE 3. True and obtained Pareto front for ZDT benchmarks.

$$P_f = \{F(xE), xE \in P_s\}$$

D. PARETO OPTIMAL FRONT (POF)

A Pareto front (P_f) illustrates the Pareto optimal set in the objective space as shown in Fig. 2. According to the above definitions, it can be expressed as Eq. (9).

As illustrated in Fig. 2, a Pareto front (P_f) depicts the Pareto optimum subset in the goal space. It may be stated as Eq. as per preceding Eq. (9).

FIGURE 4. True and obtained Pareto front for DTLZ2 and DTLZ4.

To solve a multi-objective problem, one must determine the Pareto optimal set, a collection of solutions representing the best equilibrium between goals.

III. MULTI-OBJECTIVE ARITHMETIC OPTIMIZATION

ALGORITHM

In the following Section, the conventional AOA method is presented first. After that, a novel multi-objective MAOA is developed and proposed for addressing multi-objective optimization s.

A. ARITHMETIC OPTIMIZATION ALGORITHM

Based on mathematical formulas and procedures, Abualigah et al. [40] suggested this approach in 2020. Like other metaheuristic algorithms, the AOA method starts with a set of random values. Each iteration determines the objective value of each solution. Two governing variables in this method, MOA and MOP, need to be adjusted before updating the position of solutions:

MOA(t) = min + t * ((Max - Min) / T)

where t is the latest iteration, T represents the maximum iterative process, and Max/Min represents the maximum and lowest values used to bind MOA.

MOP(t) = 1 - (t / alpha)

where alpha is a controlling parameter.

A nonrandom number called r1 is created after MOA and MOP have been updated to shift between exploration and exploitation. The exploration search processes are used to find the available solutions and improve them according to the mathematical of AOA. Additionally, during this improvement stage, the exploration operators (D and M) were used to help the exploitation step of the search phase by enhancing communication between both. The following equation is used for exploration: xi,j (t + 1)

(best xj - LBj) * MOP + UBj * mu + LBj =
if r2 < 0.5
||| (best xj - LBj) * MOP + UBj * mu + LBj =
(12)

— where mu is a parameter, is a small number, and r2 is a random value. Best(xj) is the best jth value in the bestobtained solution so far. UBj and LBj are the

TABLE 7. The statistical results of engineering problems for GD performance metric.

Functions		Algorithm			
		MOPSO	MSSA	MOALO	MAOA
BNH	Ave	3.3894E-02	6.4575E-02	5.3758E-02	1.5266E-01
	SD	2.1392E-03	1.5035E-02	1.7132E-02	6.0530E-02
CONSTR	WRT	-	+	+	
	Ave	8.3098E-04	1.6878E-03	1.5711E-03	6.7091E-04
DISK BREAK	SD	3.3369E-04	5.8950E-04	7.7806E-04	1.9336E-04
	WRT	+	+	+	
	Ave	2.3045E-03	6.2545E-03	4.1887E-02	2.1020E-03
	SD	5.6273E-04	3.3423E-03	5.4951E-03	5.3082E-04
4-BAR TRUSS	WRT	+	+	+	
	Ave	1.4095E+01	7.7966E+00	2.5901E+00	2.1487E+01
	SD	5.1580E-01	3.5486E+00	2.2161E+00	1.7221E+00
WELDED BEAM	WRT	-	+	+	
	Ave	1.0846E-02	6.3467E-03	4.4399E-03	4.2899E-03
	SD	1.9956E-03	3.0983E-03	6.9956E-04	1.3605E-03
OSY	WRT	+	+	-	
	Ave	3.5765E+00	9.4637E-01	7.0760E-01	1.9774E+00
	SD	2.5250E+00	1.9612E-01	2.4932E-01	1.5031E+00
SPEED REDUCER	WRT	+	-	+	
	Ave	1.8394E+01	7.8657E+00	7.6817E+00	6.8854E+00
	SD	5.3101E+00	3.6189E+00	3.4697E+00	2.9191E+00
SRN	WRT	+	+	+	
	Ave	3.1617E-02	4.1566E-02	1.5798E-02	4.8994E-01
	SD	1.0695E-02	2.4971E-02	3.2892E-03	1.3375E-01
	WRT	+	+	-	
VOLUME 10, 2022	W+/W-	12/4	16/0	12/4	41147
	+/-/=	6/2/0	8/0/0	6/2/0	

lower and upper boundaries of the underlying problem.

The exploitation search processes of the AOA are used to search for a new solution according to the available candidate solutions. AOA's exploitation techniques (S and A) investigate the search area carefully in many dense places and use a two-pronged approach to find a better solution. The following equation is utilized for the exploitation process:

$$\begin{aligned}
& x_{ij}(t+1) \\
& \quad \left(\text{best}_{xj} - MOP \times UB_j - LB_j \right) \times \mu + LB_j \\
& \quad \left| \left| \left| \right. \right. \right\} \text{ if } r3 < 0.5 \\
& = \\
& \quad \left(\text{best}_{xj} + \left(MOP \times UB_j - LB_j \right) \right) \\
& \quad \left| \left| \left| \right. \right. \right\} \text{ best if } r3 \geq 0.5 \times UB_j \times \mu + LB_j
\end{aligned} \tag{13}$$

B. ARCHIVE-BASED MULTI-OBJECTIVE ARITHMETIC OPTIMIZATION ALGORITHM (MAOA)

In order to execute multi-objective optimization, AOA has received two further upgrades. The functions utilized are identical to those in MOPSO [41]. The first is an archive containing all of the

TABLE 8. The statistical results of engineering problems for IGD performance metric.

non-dominated Pareto optimum solutions thus far. The following function is the leader selection feature, which aids in selecting the best available place solutions from the repository as the search process's leaders.

The archive is a simple storing mechanism for nondominated Pareto optimum solutions produced up to this point. Because it controls the archive when a solution is meant to be archived or when the archive is complete, an archive controller is a vital component. Consider that only a tiny number of people can contribute to the archive. Nondominated solutions produced up to this point are compared against the existing archive during the training process of an iteration. Three different situations might happen:

- The search agents trade details about the search space, rapidly moving toward the genuine Pareto optimal front.
- Multi-objective strategies aid in obtaining a single run that approximates the whole genuine Pareto optimal front.
- Keeping a problem's multi-objective solution allows for problem investigation across various design requirements and operational environments.

Functions		Algorithm			
		MOPSO	MSSA	MOALO	MAOA
BNH	Ave	9.6868E-03	3.3817E-03	7.5631E-03	1.0889E-03
	SD	1.7147E-04	1.1513E-03	4.5344E-03	1.4497E-04
	WRT	+	+	+	
CONSTR	Ave	5.1838E-04	7.4460E-04	2.1706E-03	2.9843E-03
	SD	5.5618E-05	1.7697E-04	7.3799E-04	1.7713E-03
	WRT	-	+	+	
DISK BRAKE	Ave	5.8831E-04	7.2091E-04	1.7399E-03	6.5336E-04
	SD	5.5995E-05	1.1206E-04	7.7421E-04	9.1198E-05
	WRT	-	+	+	
4-BAR TRUSS	Ave	2.0410E-02	2.1409E-02	2.1434E-02	2.0225E-02
	SD	3.9632E-05	3.5445E-04	3.8292E-04	2.6571E-04
	WRT	+	+	+	
WELDED BEAM	Ave	5.9705E-04	1.3441E-03	4.8242E-03	1.0993E-03
	SD	4.6341E-05	3.9377E-04	3.5852E-03	2.5765E-04
	WRT	-	+	+	
OSY	Ave	1.4663E-02	7.6220E-03	7.7532E-03	7.1949E-03
	SD	8.6917E-03	6.4006E-04	1.2021E-03	2.8459E-03
	WRT	+	-	+	
SPEED REDUCER	Ave	6.6530E-02	1.4243E-02	1.4933E-02	1.3731E-02
	SD	3.1310E-02	3.2032E-03	5.7981E-03	9.1710E-03
	WRT	+	-	+	
SRN	Ave	4.5146E-04	2.4823E-03	2.2308E-03	6.4963E-04
	SD	1.1656E-04	1.0614E-03	1.5266E-03	2.3927E-04
	WRT	-	+	+	
w^+/w^-		10/6	16/0	16/0	
+/-/=		5/3/0	8/0/0	8/0/0	

The likelihood of removing an answer increases directly to the number of responses in the hypercube (Section). If the archive is complete, one of the most densely inhabited areas is chosen for elimination solutions first. In order to create room for the new solution, a solution is randomly eliminated from one of them.

When a solution is found outside the hypercubes, a unique scenario arises. As a result, different alternative solutions' components can be modified.

According to the second feature, the leader selection function in MAOA, the best answers obtained thus far are used as the best position. This function leads the other search agents to exciting parts of the search space to find a solution close to the global optimum. However, because of the Pareto optimality ideas discussed in the preceding subsection, the answers in a multi-objective search space cannot be quickly compared. The leader selection function was created to address this problem. As previously mentioned, there is a repository of the best non-dominated answers gathered thus far. The leader selection part selects the least-crowded sections of the search space and supplies among its non-dominated answers. For each hypercube, a roulette-wheel method determines the hypercube with the following probability:

$$P_i = \frac{C}{N_i}$$

In this part, a constant integer greater than one is the number of acquired Pareto optimum responses. Eq. (14) shows that hypercubes with a lower population are more likely to recommend new leaders. The probability of selecting a hypercube from which to select leaders grows as the hypercube's range of discovered solutions decreases.

The MAOA algorithm gets its convergence from the AOA algorithm. Suppose we pick one of the archiving solutions. In that case, the AOA algorithm will almost surely be able to improve on its already great consistency. On the other hand, finding the Pareto optimum responses created a huge variety. This issue was solved by combining the leader function collection with archive maintenance. Regarding computational complexity, when n is the population's total number of individuals and m is the total number of objectives, the computing complexity of MAOA is $O(mn^2)$. The computational complexity is superior to techniques that have $O(mn^3)$ complexity. We employed the leader function collection and

TABLE 9. The statistical results of engineering problems for MS performance metric.

Functions		Algorithm			
		MOPSO	MSSA	MOALO	MAOA
BNH	Ave	1.0000E+00	7.6222E-01	5.4090E-01	1.0000E+00
	SD	0.0000E+00	1.3378E-01	1.0692E-01	0.0000E+00
	WRT	=	+	+	
CONSTR	Ave	9.9384E-01	9.0536E-01	7.7822E-01	8.7677E-01
	SD	6.8603E-03	4.8380E-02	7.6343E-02	9.9189E-02
	WRT	-	+	+	
DISK BRAKE	Ave	9.9928E-01	7.9510E-01	9.1779E-01	9.9981E-01
	SD	1.1417E-03	1.3013E-01	2.0904E-01	5.7629E-03
	WRT	-	+	+	
4-BAR TRUSS	Ave	1.4876E+00	1.2019E+00	8.4793E-01	1.4879E+00
	SD	5.4502E-04	1.8532E-01	1.2454E-01	2.3406E-06
WELDED BEAM	WRT	+	+	+	
	Ave	1.0072E+00	7.9085E-01	6.2463E-01	1.0216E+00
	SD	6.1053E-02	1.4645E-01	6.9354E-02	5.8530E-02
OSY	WRT	+	+	+	
	Ave	3.2390E-01	6.2048E-01	5.7530E-01	8.2944E-01
	SD	3.4284E-01	8.0930E-02	2.6351E-02	3.5814E-01
SPEED REDUCER	WRT	+	+	-	
	Ave	1.0000E-05	7.2004E-01	6.6868E-01	7.5636E-01
	SD	2.0000E-04	6.9034E-02	6.0939E-02	4.0006E-02
SRN	WRT	+	-	+	
	Ave	9.0900E-01	7.0583E-01	3.9177E-01	9.3221E-01
	SD	4.5463E-02	1.5109E-01	8.2165E-02	6.4793E-02
	WRT	-	+	+	
W ⁺ /W ⁻		12/2	16/0	16/0	
+/-/=		6/1/1	8/0/0	8/0/0	

TABLE 10. The statistical results of engineering problems for S performance metric.

Functions		Algorithm			
		MOPSO	MSSA	MOALO	MAOA
BNH	Ave	1.0901E+00	1.2546E+00	8.8402E-01	1.0196E+00
	SD	1.3631E-01	3.9410E-01	4.4648E-01	6.7887E-01
	WRT	-	+	+	
CONSTR	Ave	5.8740E-02	5.5908E-02	6.9071E-02	5.1010E-02
	SD	5.8936E-03	1.4886E-02	1.8623E-02	1.2462E-02
	WRT	-	+	+	
DISK BRAKE	Ave	1.2452E-01	1.2768E-01	1.4457E-01	1.1983E-01
	SD	1.3022E-02	7.0776E-02	1.1754E-02	7.5928E-03
	WRT	+	+	+	
4-BAR TRUSS WELDED BEAM	Ave	5.3605E+00	6.1160E+00	4.6988E+00	9.3337E+00
	SD	2.6169E-01	1.6605E+00	1.1721E+00	1.9859E+00
	WRT	-	+	+	
OSY	Ave	2.3432E-01	1.8912E-01	2.2431E-01	2.0362E-01
	SD	2.5702E-02	8.7588E-02	1.3595E-01	2.4473E-02
	WRT	+	+	+	
SPEED REDUCER	Ave	1.1278E+00	1.3502E+00	1.4382E+00	4.6936E+00
	SD	1.4675E+00	6.2967E-01	4.8498E-01	3.6191E+00
	WRT	+	+	-	
SRN	Ave	3.1045E+02	1.3952E+01	3.6722E+01	4.7177E+01
	SD	3.0044E+00	9.8199E+00	9.3623E+00	1.0700E+00
	WRT	+	+	+	
	Ave	2.2396E+00	2.2721E+00	1.7386E+00	2.2164E+00
	SD	4.7324E-01	9.3573E-01	8.6872E-01	1.2625E+00
	WRT	-	+	+	
W^+/W^-		14/2	12/4	12/4	
$+/-/=$		7/1/0	6/2/0	6/2/0	

archive maintenance to solve this problem. After all, the pseudo-code for MAOA is as follows:

All of the AOA capability is acquired by the MAOA algorithm, suggesting that search agents can explore and utilize the search space similarly. The primary distinction is that AOA multi-objective preserves non-dominated services in an external database and executes searches based on a group of archive members.

IV. RESULTS AND DISCUSSION

Performance measurements and case studies, which comprise unconstrained and constrained bi- and tri-objective mathematics and real-world engineering design problems, are used to assess the efficacy of the suggested technique in this Section. Multi-objective optimizers must tackle non-convex and non-linear problems, among other characteristics, to be tested by these challenges and mathematical calculations. The algorithm was written in

MATLAB 2021a. The computer's features are configured as follows: The CPU on a Macintosh computer is 2.3 GHz (an Intel Core i9 computing

platform), with 16 GB 2400 MHz DDR4 RAM (MacOS Monterey).

A. PERFORMANCE METRICS

To evaluate the algorithms' results, the following four metrics are used [42], [43]:

1) GENERATIONAL DISTANCE (GD)

The total of the distances between solution candidates produced using several approaches is an intelligent indicator for analyzing the convergence characteristics of meta-heuristic algorithms with multiple objectives.

$$GD = \left(\frac{1}{n_{pf}} \sum_{i=1}^{n_{pf}} dis_i^2 \right)^{\frac{1}{2}}$$

2) SPACING (S)

A measure to show how far candidates are from each other when compared to different sets of results achieved by multiple algorithms.

$$S = \frac{1}{n_{pf}} \sum_{i=1}^{n_{pf}} (d_i - d)^{\frac{1}{2}} \quad \text{Where } d = \frac{1}{n_{pf}} \sum_{i=1}^{n_{pf}} d_i \quad (16)$$

3) MAXIMUM SPREAD (MS)

It depicts the distribution of candidates among other obtained sets by considering the various optimal choices.

$$MS = \frac{1}{m} \sum_i \left[\frac{\max(f_i, F_i) - \min(f_i, F_i)}{F_i^{\max} - F_i^{\min}} \right]$$

4) INVERTED GENERATIONAL DISTANCE (IGD)

Multiple objective optimization techniques can be used to accurately estimate the performance of Pareto front approximations by this measure [44].

$$IGD = \frac{\sum_{i=1}^n d_i^2}{n}$$

Wilcoxon's Rank-Sum Test (WRT): In order to evaluate whether two or more datasets are from the same dispersed population, Wilcoxon's rank-sum test is a non-parametric statistical test with a 5% significance level. The performance of the algorithm is thoroughly assessed using the Wilcoxon rank-sum test. According to the null hypothesis, the two algorithms have no difference in performance because their mean metrics are identical. According to the alternative hypothesis, the mean metrics obtained by the two examined algorithms are different. The symbols “-,” “+,” and “=” are used to compare the algorithms suggested in this article to other algorithms. The algorithm performs poorly, substantially better than that, and there is no visible difference between the two.

The IGD and GD performance measures quantify the convergence, and the S and MS

of Pareto optimal is shown. Table 1 summarizes all of the listed algorithms' initial parameters. It is

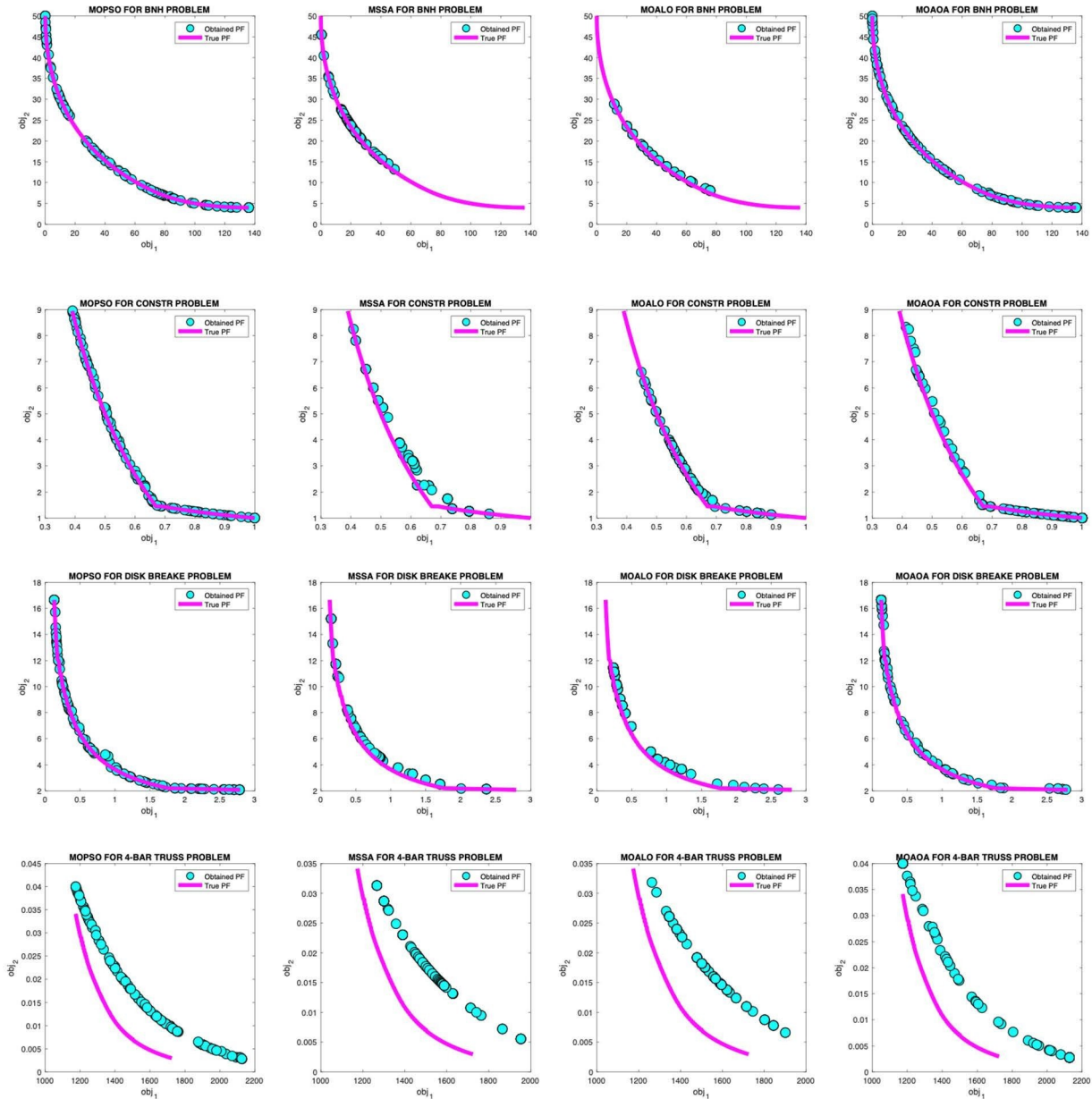


FIGURE 5. True and obtained Pareto front for engineering design problems (BNH, CONSTR, DISK BRAKE and 4-BAR TRUSS).

measure the coverage of Pareto optimum solutions predicted by the algorithms. The Wilcoxon rank-sum test was used to assess algorithm performance based on the mean value, and it successfully illustrated the algorithm's high degree of competitiveness and effectiveness.

B. EXPERIMENTAL SETUP

The MOPSO [41], MSSA [45], MOALO [34], NSGA2 [46] and MOGWO [47] are compared to MAOA in this Section, and the best figure of a set

worth noting that 100 populations and a maximum of 1000 iterations were used for each experiment. This Section tests the efficiency of the proposed algorithm on twenty-five experiments, including seven unconstrained, constrained functions, eight engineering design problems, and ten CEC-2020 mathematic functions.

The proposed MAOA's efficiency is evaluated using ZDT, DTLZ, and eight multi-objective engineering design problems. Each of the five evaluation problems in the ZDT benchmark has

two goals as shown in Table 2. DTLZ comprises two test problems with three goals and twelve variables for each function, as shown in Table 2. Engineering design problems contain eight multi-objective optimization problems; each problem has several variables given in Appendix A.

the generational distance performance metric. It is clear that the proposed MAOA has better results compared to other comparative methods (MOPSO, MSSA, and MOALO). The proposed MAOA acquired the most excellent results in four out of seven test cases (i.e., ZDT1, ZDT2, ZDT4, and

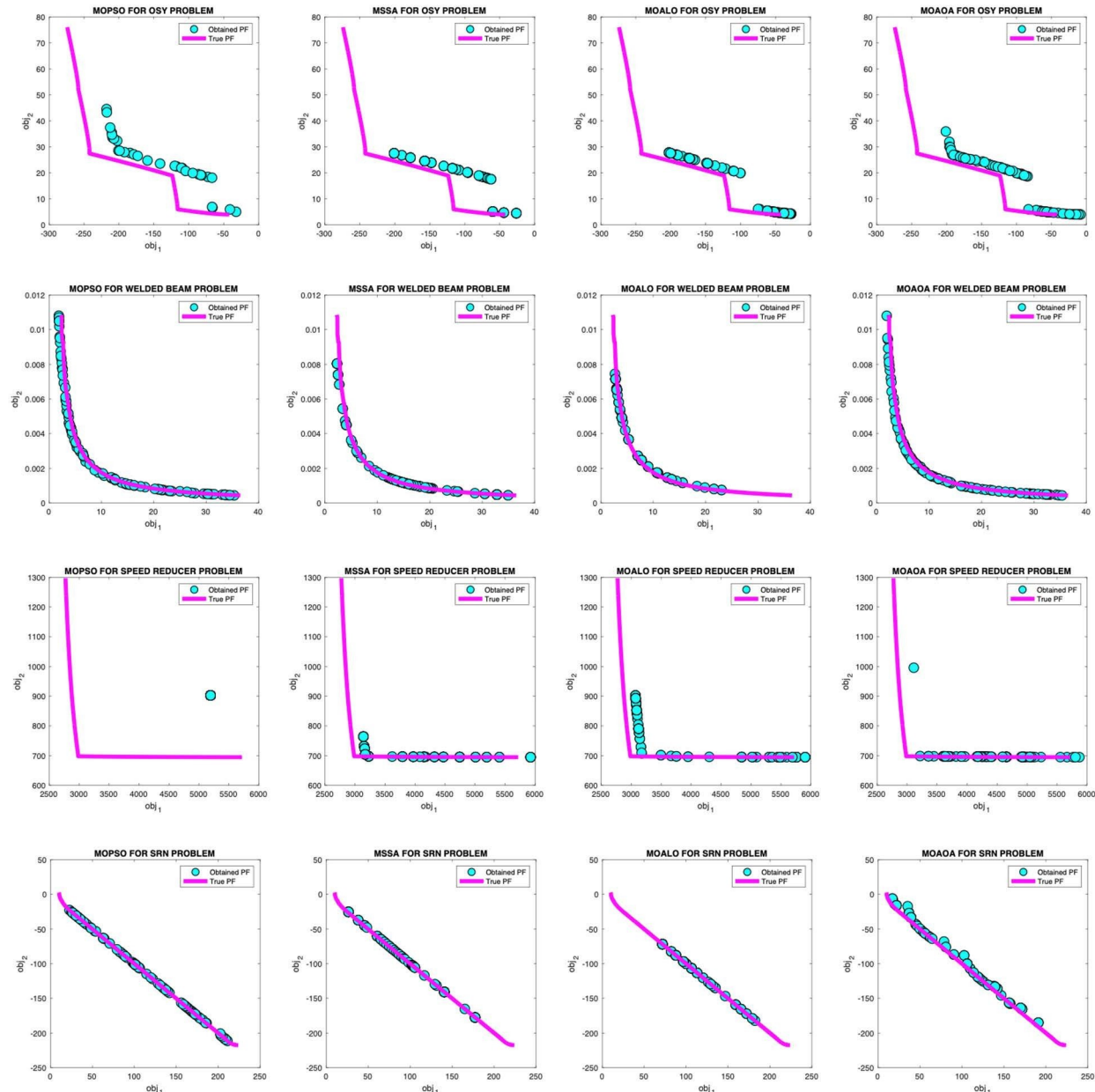


FIGURE 6. True and obtained Pareto front for engineering design problems (WELDED BEAM, OSY, SPEED REDUCER and SRN).

These benchmarks include the most recent and complex test functions: convex, non-convex, and disconnected Pareto ideal forms.

C. DISCUSSION OF THE MATHEMATICAL TEST FUNCTIONS

Table 3 shows the statistical results of mathematical functions (i.e., ZDT and DTLZ) for

ZDT6), followed by MOPSO, which brought the best results in three out of seven test cases (i.e., ZDT3, DTLZ2, and DTLZ4). Thus, the results illustrated the ability of the proposed MAOA to solve complex mathematical multi-objective optimization problems. It acquired excellent results in most problems tested with low SD values.

Table 4 shows the statistical results of mathematical functions (i.e., ZDT and DTLZ) for the inverted generational distance performance metric. The proposed MAOA has better results than other comparative methods (MOPSO, MSSA, and MOALO). The proposed MAOA acquired the most excellent results in five out of seven test cases (i.e.,

TABLE 11.

GD.

Function		Algorithm					
		MOPSO	MSSA	MOALO	NSGA2	MOGWO	MAOA
M1: MMF1	Ave	1.5852E-03	5.5321E-03	3.8109E-03	3.7211E-02	3.7425E-03	1.4005E-03
	SD	1.1756E-03	4.0271E-03	2.3997E-03	2.8343E-02	4.3844E-03	4.4849E-04
	WRT	+	+	+	+	+	
M2: MMF2	Ave	8.3130E-04	4.0660E-02	7.8547E-02	9.0907E-02	4.1874E-04	2.5421E-03
	SD	2.0555E-04	7.4608E-02	1.4097E-01	1.7739E-03	7.4551E-05	3.4512E-04
	WRT	+	+	+	+	-	
M3: MMF4	Ave	5.5744E-03	2.4294E-03	6.1889E-03	2.8266E-03	2.6278E-03	2.3245E-03
	SD	7.4168E-03	2.0292E-03	9.3279E-03	3.6950E-03	3.2182E-03	1.3201E-03
	WRT	+	+	+	+	+	
M4: MMF5	Ave	5.4699E-03	2.7860E-02	3.9652E-03	2.0062E-03	8.1579E-03	1.9954E-03
	SD	2.4279E-03	2.2339E-02	3.8404E-03	2.3322E-03	9.2404E-03	1.0019E-03
	WRT	+	+	+	+	+	
M5: MMF7	Ave	1.3656E-03	1.8240E-03	1.4219E-03	3.6400E-03	1.4260E-03	2.8795E-03
	SD	2.6484E-03	6.7194E-04	7.8765E-04	5.2526E-03	5.4680E-04	2.8741E-04
	WRT	-	+	+	+	+	
M6: MMF8	Ave	6.5325E-04	2.1142E-01	8.0254E-04	9.5964E-03	3.4481E-02	3.3295E-03
	SD	4.1323E-05	3.0208E-01	6.2608E-04	3.5307E-03	7.5762E-02	1.4044E-03
	WRT	-	+	+	+	+	
M7: MMF10	Ave	3.6524E-03	2.3929E-02	8.0650E-03	2.4629E-03	3.3425E-03	7.1412E-03
	SD	3.8649E-04	4.9091E-02	1.0934E-02	1.4128E-02	1.7242E-03	2.0019E-03
	WRT	+	+	+	-	+	
M8: MMF11	Ave	5.2576E-03	3.0179E-03	2.2348E-03	1.1193E-02	2.9190E-03	2.0094E-03
	SD	2.6945E-04	1.8586E-03	1.1111E-03	8.2984E-01	6.2066E-04	2.2841E-04
	WRT	+	+	+	+	+	
M9: MMF12	Ave	5.6761E-04	5.0576E-04	5.1757E-04	4.4113E-03	4.2095E-04	3.9825E-04
	SD	6.1558E-05	4.7320E-05	1.3891E-04	2.8398E-03	5.9355E-05	4.3251E-05
	WRT	+	+	+	+	+	
M10: MMF13	Ave	2.1242E-03	1.8325E-03	1.7252E-03	7.8191E-03	1.8095E-03	1.2341E-03
	SD	1.9934E-04	8.0983E-04	1.3299E-03	1.5743E-04	4.4849E-04	1.4254E-04
	WRT	+	+	+	+	+	
W^+/W^- /-/=		16/4	20/0	20/0	18/2	18/2	
		8/2/0	10/0/0	10/0/0	9/1/0	9/1/0	

ZDT1-ZDT6), followed by MOPSO, which provided the best results in two out of seven test cases (i.e., DTLZ2 and DTLZ4). Thus, the results illustrated the ability of the proposed MAOA to solve complex mathematical multi-objective optimization problems. The proposed MAOA produced the best results in most problems tested with low SD values. The other comparative methods (i.e., MSSA and MOALO) did not obtain any best results regarding the inverted generational distance performance metric.

Table 5 shows the statistical results of mathematical functions (i.e., ZDT and DTLZ)

for the maximum spread performance metric. The proposed MAOA has better results than other comparative methods (MOPSO, MSSA, and MOALO) in almost all the tested cases. The proposed MAOA acquired the most excellent results in seven out of seven test cases (i.e.,

ZDT1-ZDT6, DTLZ2, and DTLZ4). The proposed MAOA produced the best results in most problems tested with low SD values, except ZDT3 and DTLZ2. The other comparative methods (i.e., MOPSO, MSSA, and MOALO) did not obtain any best results regarding the

maximum spread performance metric.

Table 6 shows the statistical results of mathematical functions (i.e., ZDT and DTLZ) for the spacing performance metric. The proposed MAOA has better results than other comparative methods (MOPSO, MSSA, and MOALO). The proposed MAOA acquired the most excellent results in three out of seven test cases (i.e., ZDT2, ZDT3, and ZDT6), followed by MSSA, which acquired the most excellent results in three out of seven test cases (i.e., ZDT4, DTLZ2, and DTLZ4), and MOALO acquired the most excellent results in one out of seven test cases (i.e., ZDT1). Thus, the results illustrated the proposed MAOA's ability to solve complex mathematical multi-objective

optimization problems. The proposed MAOA produced the best results in most problems tested with low SD values, except ZDT1, DTLZ2, and DTLZ4. The MOPSO does not obtain any best results regarding the spacing performance metric.

Figs. 3 and 4 show the true and obtained Pareto front for ZDT and DTLZ benchmark problems for all the comparative methods. The Pareto front results of the proposed MAOA and other comparative methods are seen in this figure, with MAOA achieving greater convergence and coverage than the different comparative algorithms. Furthermore, MAOA's solutions are widely spread around the entire Pareto front. Where MSSA gives a non-uniform distribution of solutions at the curve's ends, the proposed

TABLE 12. IGD.

Function		Algorithm					
		MOPSO	MSSA	MOALO	NSGA2	MOGWO	MAOA
M1: MMF1	Ave	9.2459E-04	4.0338E-03	3.4017E-03	6.3276E-04	2.7547E-03	9.0124E-04
	SD	1.2258E-04	7.8337E-04	4.0413E-04	1.1765E-04	2.3995E-04	3.6524E-04
	WRT	+	+	+	-	+	
M2: MMF2	Ave	2.2238E-03	4.4213E-03	2.2902E-03	9.2533E-03	9.4570E-04	2.5741E-03
	SD	8.4385E-04	1.5387E-03	6.1735E-04	1.7930E-03	1.1935E-04	5.6866E-04
	WRT	+	+	+	+	-	
M3: MMF4	Ave	8.8933E-04	4.9496E-03	4.8819E-03	4.3677E-04	1.7148E-03	3.6225E-04
	SD	1.2397E-04	2.5771E-03	1.7092E-03	2.4241E-05	3.3043E-04	8.7402E-04
	WRT	+	+	+	+	+	
M4: MMF5	Ave	8.7820E-04	3.3427E-03	2.1757E-03	8.0141E-04	2.2009E-03	7.6001E-04
	SD	9.3769E-04	6.0112E-04	1.7789E-04	4.1688E-04	1.0885E-03	9.8527E-05
	WRT	+	+	+	+	+	
M5: MMF7	Ave	5.5406E-04	4.2735E-03	7.6473E-03	4.1358E-04	1.9071E-03	1.1013E-03
	SD	1.2011E-04	1.8876E-03	5.8726E-03	9.6031E-05	7.7299E-05	5.3301E-03
	WRT	+	+	+	-	+	
M6: MMF8	Ave	4.6497E-04	2.2485E-03	2.0296E-03	1.6972E-02	9.5161E-04	2.3001E-03
	SD	1.7000E-05	4.2443E-04	5.4512E-04	1.2889E-02	4.2348E-04	2.3341E-04
	WRT	-	+	+	+	+	
M7: MMF10	Ave	8.8248E-04	1.0774E-02	1.0450E-02	1.0269E-01	1.3264E-03	1.0122E-03
	SD	1.2160E-04	6.9750E-03	5.0625E-03	7.2147E-03	4.3239E-04	4.4717E-03
	WRT	-	+	+	+	+	
M8: MMF11	Ave	8.3217E-04	8.1063E-03	1.3618E-02	9.3848E-02	4.3671E-03	6.7458E-04
	SD	1.2475E-04	4.8219E-03	3.8071E-03	1.0381E-03	3.1204E-03	1.1017E-04
	WRT	+	+	+	+	+	
M9: MMF12	Ave	5.2314E-04	6.6534E-03	6.6576E-03	7.1622E-03	1.5235E-03	4.9854E-04
	SD	3.9743E-05	7.1555E-03	2.7721E-03	7.6951E-03	5.6604E-04	3.4307E-04
	WRT	+	+	+	+	+	
M10: MMF13	Ave	4.4906E-04	4.5576E-03	9.0777E-03	5.8677E-02	1.9444E-03	3.9321E-04
	SD	7.0224E-04	2.1694E-03	5.8111E-03	1.8127E-02	6.5347E-04	5.5865E-04
	WRT	+	+	+	+	+	
W+/W-		16/4	20/0	20/0	16/4	18/2	
41156 /-/=		8/2/0	10/0/0	10/0/0	VOLUME 10, 2023 8/2/0	9/1/0	

MAOA algorithm is superior. MAOA generated more convergent and widely spread solutions on the real Pareto optimal. Other comparative methods could not cover any of the true Pareto optimal solutions for these functions, but they could cover one end of the true Pareto. Simultaneously, it crashes on the other end. Based on the Wilcoxon rank sum test findings shown in Tables 3 to 6, it is clear that the MAOA is significantly more efficient than the other methods.

D. DISCUSSION OF THE ENGINEERING PROBLEMS

In this Section, the proposed MAOA method is employed to solve several different multi-objective engineering design problems and compared with other well-known multiobjective methods.

Table 7 shows the statistical results of the tested engineering problems for the generational distance performance metric. The proposed MAOA got promising results for all the tested problems. MAOA acquired the most excellent results for four out of eight tested problems: CONSTR, DISK BRAKE, WELDED BEAM, and SPEED REDUCER problems. Followed by MOPSO, which acquired the most excellent results on two out of eight test problems (i.e., BNH and 4-BAR TRUSS), and MOALO acquired the most excellent results on two out of eight test problems (i.e., OSY and SRN). Moreover, the proposed MAOA got stable results according to the SD measure, which reflects the ability of the proposed MAOA over several runs. We conclude from the obtained results that the proposed MAOA method has achieved stimulating results and

has a high and balanced capacity to solve complex engineering problems with multiple objectives.

Table 8 presents the statistical results of the tested engineering problems for the inverted generational distance performance metric. The proposed MAOA produced encouraging results for all the tested problems. MAOA acquired the best results for five out of eight tested problems: BNH, DISK BRAKE, 4-BAR TRUSS, OSY, and SPEED REDUCER.

Followed by MOPSO, it obtained the best results on three out of eight test problems (i.e., CONSTR, WELDED BEAM, and SRN). Besides, the problems for the maximum spread performance metric. The proposed MAOA got better outcomes for all the tested problems compared to other

TABLE 13.

MS.

Function		Algorithm					
		MOPSO	MSSA	MOALO	NSGA2	MOGWO	MAOA
M1: MMF1	Ave	1.0405E+00	1.0292E+00	9.6095E-01	1.0008E+00	1.1469E+00	1.1889E+00
	SD	3.8703E-02	7.4761E-02	7.9028E-02	4.6539E-01	2.8675E-01	3.6523E-02
	WRT	+	+	+	+	+	
M2: MMF2	Ave	1.0000E+00	1.5837E+00	2.4489E+00	4.8560E-01	9.9776E-01	8.9984E-01
	SD	0.0000E+00	1.1692E-01	2.1803E-01	9.1533E-01	5.0063E-03	1.5142E-02
	WRT	+	+	-	+	+	
M3: MMF4	Ave	1.1564E+00	8.6683E-01	8.9243E-01	7.1184E-01	1.1074E+00	1.2006E+00
	SD	2.3744E-01	8.7815E-02	9.4367E-02	1.4056E-01	1.8442E-01	6.3253E-02
	WRT	+	+	+	+	+	
M4: MMF5	Ave	1.1281E+00	1.0736E+00	1.0848E+00	1.0893E+00	1.2667E+00	9.9851E-01
	SD	5.3298E-02	4.3644E-01	1.1268E-01	1.7275E-01	2.1784E-01	4.6521E-02
	WRT	+	+	+	+	-	
M5: MMF7	Ave	1.0621E+00	7.5755E-01	6.7909E-01	1.1233E+00	9.5926E-01	9.3541E-01
	SD	6.0923E-02	9.0877E-02	1.8167E-01	2.0187E-01	1.3433E-02	1.0141E-02
	WRT	+	+	+	-	+	
M6: MMF8	Ave	9.9669E-01	1.5597E+00	9.3247E-01	1.3965E+00	2.9770E+00	9.7548E-01
	SD	3.6031E-03	1.0455E+01	6.3542E-02	1.5714E-01	9.5271E-01	2.2541E-02
	WRT	+	+	+	+	-	
M7: MMF10	Ave	1.0000E+00	8.5590E-01	7.6156E-01	6.5685E-01	9.8906E-01	7.2518E-01
	SD	7.8300E-03	2.2228E-01	1.2705E-01	1.0628E-01	3.4240E-02	9.3253E-03
	WRT	-	+	+	+	+	
M8: MMF11	Ave	9.6992E-01	7.4196E-01	5.1036E-01	2.7847E-02	8.7184E-01	9.8516E-01
	SD	2.3850E-02	8.4684E-02	1.1371E-01	2.6141E-02	7.4918E-02	1.0393E-02
	WRT	+	+	+	+	+	
M9: MMF12	Ave	9.0156E-01	8.4380E-01	8.4393E-01	9.1950E-01	9.7502E-01	9.8369E-01
	SD	1.2421E-03	2.1330E-01	6.3666E-02	1.4823E-01	2.3631E-02	1.0258E-02
	WRT	+	+	+	+	+	
M10: MMF13	Ave	9.9921E-01	7.5131E-01	5.2827E-01	6.1534E-01	8.7876E-01	9.9976E-01
	SD	3.0561E-02	1.2026E-01	2.1729E-01	9.3616E-02	4.0393E-02	3.0393E-02
	WRT	+	+	+	+	+	
W^+/W^- /-/=		18/2	20/0	18/2	18/2	16/4	
		9/1/0	10/0/0	9/1/0	9/1/0	8/2/0	

proposed MAOA got well-built results according to the SD measure, which reflects the ability of the proposed MAOA over various runs. From these results, we concluded that the proposed MAOA method has reached exciting developments and has a high and stable capability to tackle complex engineering problems with multi-objective functions.

Table 9 shows the results of the tested industrial engineering

comparative methods. MAOA acquired the most excellent results on seven out of eight tested problems. Followed by MOPSO, it acquired the most excellent results on two out of eight test problems (i.e., BNH and CONSTR).

Table 10 shows the statistical results of the tested engineering problems for the spacing performance metric. The proposed MAOA got better results for all the tested

problems compared to other multi-objective methods. MAOA acquired the most excellent results on three out of eight tested problems: BNH, CONSTR, and DISK BRAKE problems. Followed by MOALO, it acquired the most excellent results on two out of eight test problems (i.e., 4-BAR TRUSS and SRN), MSSA acquired the most excellent results on two out of eight test problems (i.e., WELDED BEAM and SPEED REDUCER), and finally, MOPSO acquired the most excellent results on one out of eight test problems (i.e., OSY). Moreover, the proposed MAOA got stable results according to the SD measure, which reflects the ability of the proposed MAOA over several runs. Finally, the proposed MAOA method has proven its effectiveness in solving various complex engineering problems.

Figs. 5 and 6 show the true and obtained Pareto front for eight engineering design problems (i.e., BNH, DISK BRAKE, 4-BAR TRUSS, WELDED BEAM, OSY, SPEED REDUCER, SRN, and CONSTR) using all the comparative methods. The achieved Pareto front results of the proposed MAOA method and other comparative methods are viewed in this figure, with MAOA achieving greater convergence and coverage than the different comparative methods. The proposed MAOA Pareto solutions are very similar to Pareto solutions to engineering design problems, as shown in this figure. The MAOA gives better results for almost all the tested engineering problems.

Figs. 5 and 6 show the true and obtained Pareto front for eight engineering design problems (i.e., BNH, DISK BRAKE, 4-BAR TRUSS,

shown in this figure. The MAOA gives better results for almost all the tested engineering problems. The findings of the Wilcoxon rank

TABLE 14.

S.

Function			Algorithm					
			MOPSO	MSSA	MOALO	NSGA2	MOGWO	MAOA
1.0405E+00	1.0292E+00	9.6095E-01	2.7041E-03	1.6387E-02	4.9232E-03	2.0626E-03	3.5578E-02	2.0087E-03
3.8703E-02	7.4761E-02	7.9028E-02	1.7504E-02	1.1864E-02	4.5731E-02	1.3061E-01	1.2108E-02	1.3725E-01
+	+	+	+	+	+	+	+	+
1.0000E+00	1.5837E+00	2.4489E+00	3.0686E-03	8.7544E-03	4.0518E-02	2.9205E-02	1.6747E-03	3.2514E-02
0.0000E+00	1.1692E-01	2.1803E-01	1.1091E-02	1.6772E-02	1.3560E-02	6.6782E-02	2.6699E-02	5.6524E-01
+	+	+	-	+	+	+	+	+
1.1564E+00	8.6683E-01	8.9243E-01	2.0380E-03	1.1620E-02	6.2102E-03	1.1259E-03	2.0243E-02	1.0047E-03
2.3744E-01	8.7815E-02	9.4367E-02	1.1973E-02	1.1296E-02	1.3611E-02	1.3390E-02	3.5138E-02	1.1125E-02
+	+	+	+	+	+	+	+	+
1.1564E+00	8.6683E-01	8.9243E-01	6.2267E-03	6.5895E-03	1.1871E-02	7.9434E-03	1.8720E-02	5.1587E-03
2.3744E-01	8.7815E-02	9.4367E-02	1.1600E-02	1.2613E-02	9.5569E-03	1.1462E-02	1.5283E-02	2.5417E-01
+	+	+	+	+	+	+	+	+
1.1281E+00	1.0736E+00	1.0848E+00	1.6462E-03	5.2882E-03	6.1320E-03	1.4521E-03	5.1392E-03	9.8881E-03
5.3298E-02	4.3644E-01	1.1268E-01	9.2245E-03	9.9820E-03	1.5631E-02	1.0510E-01	1.7856E-01	5.8541E-02
+	+	+	+	+	+	-	+	+
1.0621E+00	7.5755E-01	6.7909E-01	3.8884E-04	3.4389E-03	4.6702E-03	6.3614E-02	3.7117E-01	1.0148E-03
6.0923E-02	9.0877E-02	1.8167E-01	4.1672E-02	3.6111E-02	5.3845E-02	7.6412E-02	8.3050E-02	4.2511E-02
+	+	+	-	+	+	+	+	+
1.0621E+00	7.5755E-01	6.7909E-01	3.7347E-03	2.1464E-02	1.9358E-02	9.0087E-02	1.9687E-02	4.9865E-02
6.0923E-02	9.0877E-02	1.8167E-01	6.6300E-02	1.0746E-01	4.7241E-02	3.2719E-01	1.1160E-01	2.0015E-01
+	+	+	-	+	+	+	+	+
9.9669E-01	1.5597E+00	9.3247E-01	8.8577E-03	9.8470E-02	5.8224E-03	2.7027E-02	2.5791E-02	4.3257E-03
3.6031E-03	1.0455E+01	6.3542E-02	1.1781E-02	1.1163E-02	3.8159E-03	4.1492E-02	1.3140E-02	1.0705E-03
+	+	+	+	+	+	+	+	+
1.0000E+00	8.5590E-01	7.6156E-01	1.3056E-03	6.0189E-03	4.0508E-03	1.8690E-02	6.5432E-03	1.0532E-03
7.8300E-03	2.2228E-01	1.2705E-01	8.7333E-02	1.0319E-01	7.6104E-02	7.2539E-01	1.3725E-01	6.9864E-02
+	+	+	+	+	+	+	+	+
1.0000E+00	8.5590E-01	7.6156E-01	5.5400E-03	6.2981E-02	1.7521E-02	1.5878E-02	5.1587E-02	5.1142E-03
7.8300E-03	2.2228E-01	1.2705E-01	1.7041E-03	1.6387E-02	4.9232E-03	2.0626E-03	3.5578E-02	4.1182E-03
+	+	+	+	+	+	+	+	+
W^+/W^-			14/6	20/0	20/0	18/2	20/0	
/-/=			7/3/0	10/0/0	10/0/0	9/1/0	10/0/0	

WELDED BEAM, OSY, SPEED REDUCER, SRN, and CONSTR) using all the comparative methods. The achieved Pareto front results of the proposed MAOA method, and other comparative methods are viewed in this figure, with MAOA achieving greater convergence and coverage than the different comparative methods. The proposed MAOA Pareto solutions are very similar to Pareto solutions to engineering design problems, as

sum test, which are provided in Tables 7 to 10, make it abundantly clear that the MAOA is a strategy that is significantly more successful than the other methods.

E. DISCUSSION OF THE CEC-2020 MMO TEST PROBLEMS

In this Section, the capability of the proposed multi-objective strategy, known as MAOA, to address a wide variety of issues related to CEC-2020 will be tested. The malfunction of the

MMO test, including all of their specifics, are described in [48].

Tables 11-14 show the obtained results for all the comparative methods in various evaluation measures. The average, standard deviation and Wilcoxon signed-rank test sign are calculated for each measure to explain the results clearly.

Table 11 presents the results of GD for the comparative methods. The proposed MAOA got the best results in most of the tested cases (i.e., MMF1, MMF4, MMF5, MMF11, MMF12, and MMF13). The performance of the proposed method is promising compared to the comparative methods like MOPSO, MSSA, MOALO, NSGA2, and MOGWO.

According to the Wilcoxon signed-rank test, the proposed MAOA almost overcame all the comparative methods.

Table 12 presents the results of IGD for the comparative methods. The proposed MAOA obtained the best outcomes in most of the tested cases (i.e., MMF4, MMF5, MMF11, MMF12, and MMF13). The performance of the proposed MAOA method is good compared to the comparative methods like MOPSO, MSSA, MOALO, NSGA2, and MOGWO. According to the Wilcoxon signed-rank test, the proposed MAOA almost overcame all the comparative methods. For example, the proposed MAOA overwhelms MOPSO in 8 cases, MSSA in 10 cases, MOALO in 10 cases, NSGA2 in 8 cases, and MOGWO in 9 cases.

Table 13 presents the results of MS for the comparative methods. The proposed MAOA got the best results in most of the tested cases (i.e., MMF1, MMF4, MMF11, MMF12, and MMF13). The performance of

the proposed MAOA method is sound likened to comparative methods like MOPSO, MSSA, MOALO, NSGA2, and MOGWO. The proposed MAOA almost overwhelmed all the comparative

FIGURE 7. True and obtained Pareto fronts for the CEC 2020 problems (M1-M5).

methods in the Wilcoxon signed-rank test. For example, the proposed MAOA overwhelms MOPSO in 9 cases, MSSA in 10 cases, MOALO in 9 cases, NSGA2 in 9 cases, and MOGWO in 8 cases.

Table 14 shows the results in terms of the S measure for the comparative methods. The

proposed MAOA got the best results in most of the tested cases (i.e., MMF1, MMF4, MMF5, MMF11, MMF12, and MMF13). The performance of the proposed MAOA method is sound likened to comparative methods like MOPSO, MSSA, MOALO, NSGA2, and MOGWO. The proposed

MAOA almost overwhelmed all the comparative methods in the Wilcoxon signed-rank test. For example, the proposed MAOA overwhelms MOPSO in

7 cases, MSSA in 10 cases, MOALO in 10 cases, NSGA2 in 9 cases, and MOGWO in 10 cases. From the given results, we concluded that the performance of the proposed MOAOA is excellent, and it

can get better results compared to other similar methods in solving various multi-objective problems.

Figures 7 and 8 show the true and obtained Pareto fronts for the CEC 2020 problems (M1-M5) and the true and obtained Pareto fronts for the CEC 2020 problems (M6-M13), respectively. These figures show that the obtained results

by the proposed MAOA are closer to the optimal solution and got almost the best results compared to similar methods. Moreover, the performance of the proposed MAOA is promising according to the presented curves in Figures 7 and 8.

V. CONCLUSION

In order to assist decision-makers, the field of multiobjective optimization provides solutions for a variety of complex problems with many conflicting objectives. The previously developed Arithmetic Optimization Algorithm (AOA) was modified to create the Archive-based Multiobjective Arithmetic Optimization Algorithm (MAOA) in order to address different multi-objective issues. The suggested MAOA is based on the distribution behavior of the most common mathematical arithmetic operators, such as multiplication, division, subtraction, and addition. In MAOA, non-dominated Pareto optimal solutions are discovered using an archived concept. To determine whether the suggested MAOA approach can be used to address actual engineering challenges, it is tested on eight constrained engineering design problems and seven benchmark test functions. Its performance is compared to those of the five well-known multiobjective techniques MOPSO, MSSA, MOALO, NSGA2 and MOGWO. Several metrics, including Generational Distance (GD), Inverted Generational Distance (IGD), Maximum Spread (MS), and Spacing (S), are used to assess the results. The experimental study demonstrated the

superior performance and high degree of convergence of the proposed MAOA technique compared to other existing algorithms. The MAOA gives the best outcomes in terms of computing costs when compared to other existing competing algorithms, according to a number of experimental studies in this respect. The binary version of the MAOA method will be developed for use in upcoming studies to address a variety of challenging, intricate real-world problems in various fields. Additionally, the future contribution of the suggested method can be seen in its many-objective form.

APPENDIX A CONSTRAINED

MULTI-OBJECTIVE TEST

PROBLEMS USED IN THIS PAPER

This issue [49] contains two constraints and two design variables with a convex Pareto front.

$$\text{Minimize : } f_1(x) = x_1 \quad (\text{A.1})$$

$$\text{Minimize : } f_2(x) = (1 + x_2)/x_1 \quad (\text{A.2})$$

$$\text{where : } g_1(x) = 6 - (x_2 + 9x_1) \quad (\text{A.3})$$

$$g_2(x) = 1 + (x_2 - 9x_1) \quad (\text{A.4})$$

$$0.1 \leq x_1$$

$$0 \leq x_2 \leq 5$$

A. SRN

Srinivas and Deb [50] proposed the following continuous Pareto optimum front for the following issue:

$$\text{Minimize : } f_1(x) = 2 + (x_1$$

$$- 2)^2 + (x_2 - 1)^2$$

$$(\text{A.5}) \text{ Minimize } f_2(x)$$

$$= 9x_1(x_2 - 1)^2$$

$$(\text{A.6}) \text{ where :}$$

$$(A.7) \quad g_2(x) = x_1 - 3x_2$$

$$+ 10 \quad (A.8)$$

$$- 20 \leq x_1 \leq 20,$$

$$-20 \leq x_2 \leq 20$$

B. BNH

Binh and Korn [51] provided this example for the first time as follows:

$$\text{Minimize : } f_1(x) = 4x_1^2 +$$

$$4x_2^2 \quad (A.9) \quad \text{Minimize :}$$

$$f_2(x) = (x_1 - 5)^2 + (x_2 - 5)^2$$

$$(A.10) \quad \text{where : } g_1(x) = (x_1$$

$$- 5)^2 + x_2^2 - 25 \quad (A.11)$$

$$g_2(x) = 7.7 - (x_1 - 8)^2 -$$

$$(x_2 + 3)^2 \quad (A.12)$$

$$0 \leq x_1 \leq 5,$$

$$\leq 3$$

C. OSY

It was suggested by Osyczka and Kundu [52] that there should be five separate regions. In addition, the following six restrictions and six design variables should be taken into account:

$$\text{Minimize:} \quad f$$

(

$$A.13) \quad \text{Minimize : } f_2(x)$$

$$= [25(x_1 - 2)^2 + (x_2 - 1)^2$$

$$+ (x_3 -$$

$$1)^2(x_4 - 4)^2$$

$$+ (x_5 - 1)^2]$$

$$(A.14)$$

$$\text{Where: } g_1(x) = 2 - x_1 - x_2$$

$$(A.15) \quad g_2(x) = -6 + x_1$$

$$+ x_2 \quad (A.16) \quad g_3(x) = -2 - x_1$$

$$+ x_2 \quad (A.17) \quad g_4(x) = -2 + x_1$$

$$- 3x_2 \quad (A.18) \quad g_5(x) = -4 + x_4$$

$$+ (x_3 - 3)^2 \quad (A.19) \quad g_6(x)$$

$$= 4 - x_6 - (x_5 - 3)^3$$

$$(A.20)$$

$$0 \leq x_1 \leq 10,$$

$$x_2 \leq 10, 1 \leq x_3 \leq 5$$

$$(A.21)$$

$$0 \leq x_4 \leq 6, 1 \leq$$

$$x_5 \leq 5, 0 \leq x_6 \leq 10$$

APPENDIX B CONSTRAINED MULTI-OBJECTIVE ENGINEERING PROBLEMS USED IN THIS PAPER A. THE FOUR- BAR TRUSS DESIGN PROBLEM

In this 4-bar truss issue [53], there are two objectives as structural volume (f_1) and displacement (f_2), which are considered to be minimized. This issue has four design variables (x_1 - x_4) according to the cross-sectional area of members 1, 2, 3, and 4. This problem may be mathematically formularized as below:

$$\text{Minimize : } f_1(x) = 200 \times (2 \times$$

$$x(1)) + \text{sqrt}(2 \times x(2))$$

$$+ \text{sqrt}(x(3)) +$$

$$(B.1)$$

$$\text{Minimize : } f_2(x) = 0.01 \times$$

$$- ((2 \times \text{sqrt}(2))/x(3)) +$$

$$(2/x(1))) \quad (B.2)$$

$$1 \leq x_1 \leq 3,$$

$$1.4142 \leq x_2$$

$$\leq 3$$

$$1.4142 \leq x_3 \leq 3$$

$$1 \leq x_4 \leq 3$$

B. THE WELDED BEAM DESIGN PROBLEM

Ray and Liew [54] first offered this welded beam issue with two objectives, namely the fabrication cost

(f_1) and beam deflection (f_2), which is to be reduced and four constraints. This issue has four numbers of variables, named weld thickness (x_1), clamped bar length (x_2), bar height (x_3), and bar thickness (x_4) as follows:

$$\text{Minimize : } f_1(x) = 1.10471 \times x(1)^2 \times x(2) + 0.04811 \times x(3) \times x(4)$$

(B.3)

$$\text{Minimize : } f_2(x) = 65856000 / (30 \times 10^6 \times x(4) \times x(3)^3)$$

$$\text{where : } g_1(x) = \tau - 13600$$

$$g_2(x) = \sigma - 30000$$

$$g_3(x) = x(1) - x(4)$$

$$g_4(x) = 6000 - P$$

$$0.125 \leq x_1 \leq 5, \quad 0$$

$$0.1 \leq x_3 \leq 10, \quad 0.$$

$$\text{where :}$$

$$D = \sqrt{J}$$

$$(B.9)$$

$$J$$

$$\sqrt{2}$$

$$(B.10)$$

$$= \alpha \sqrt{2} * x(1) * x(2)$$

$$D$$

$$J \quad \beta = Q * \frac{D}{J}$$

C. DISK BRAKE DESIGN PROBLEM

This disc brake design issue has been figured out by Ray and Liew [54] with two objectives, namely stopping time (f_1) and brake mass (f_2) to minimize and five

constraints for a disc brake. This problem contains five numbers of variables: disc inner radius (x_1), outer disc radius (x_2), engaging force (x_3), and friction surfaces number (x_4). The following equations may represent this problem:

$$\text{Minimize : } f_1(x) = 4.9 \times (10)^{-5} \times x(2)^2 - x(1)^2$$

$$\times x(4) - 1$$

$$(B.13)$$

$$\text{Minimize : } f_2(x) = (9.82 \times (10)^6) \times x(2)^2 - x(1)^2$$

$$\times x(2)^2 - x(1)^2) / ((x(2))^3 - x(1)^3)$$

$$\times x(4) \times x(3)$$

$$(B.14) \quad g_1(x) =$$

$$20 + x(1) -$$

$$x(2) \quad (B.15) \quad g_2$$

$$(x) = 2.5 + (x$$

$$(4) + 1) - 30$$

$$(B.16)$$

$$g$$

$$-$$

$$0.4$$

$$g_4$$

$$(x)$$

$$(B.11)$$

$$(B.12)$$

$$=$$

$$(2.$$

$$22$$

$$\times$$

$$(10$$

$$)^{-3})$$

$$\times x \quad \times (3.3333 \times x$$

$$(3)$$

$$\times \quad \times x$$

$$- 1 \quad - 43.0934) - 1.508 \times x(1) \times$$

$$g_5(x) = 900 - (2.66 \times (10)^{(-2)} \times (x(6)^2 + x(7)^2$$

$$x(3) \quad \text{B.20) Minimize : } f_2(x) =$$

$$\times x \quad ((\text{sqrt}(((745 * x(4))/x(2) * x(3))))^2$$

$$/ \quad \text{where} \quad :$$

$$\times$$

$$(B.19) \quad 55 \leq x_1 \leq 80, \quad g_2(x) = 397.5/(x$$

$$\leq x_2 \leq 110 \quad (1) \times x(2)^2 \times x$$

$$1000 \leq x_3 \leq 3000$$

$$2 \leq x_4 \leq 20$$

D. SPEED REDUCER DESIGN PROBLEM

This issue [53], [55] contains two objectives such as weight (f_1) and stress (f_2), which are to be minimized. The problem may represent with a diagram as given in Fig. 9. Also, the problem has eleven constraints with seven numbers of design variables such as the width of the gear face (x_1), teeth module (x_2), pinion teeth number (x_3 numeral variable), a distance between of bearings 1 (x_4), a distance of bearings 2 (x_5), shaft 1 diameter (x_6), and shaft 2 diameter (x_7). The equations may clearly represent this problem as below:

$$g$$

$$\times x(3) \times x \quad \text{(B.24)} \quad g$$

$$\times x(3)$$

$$\times \quad x \quad \text{(B.25)} \quad g_5$$

$$(x) = ((\text{sqrt}((745 \times x(4))/x(2)$$

$$\times x$$

$$\times x$$

$$g_6(x) = ((\text{sqrt}((745 \times x(4))/x(2)$$

$$\times x$$

$$\text{Minimize : } f_1(x) = 0.7854 \times x(1) \times x(2)^2$$

$$\times x$$

$$g_7(x) = ((x(2) \times x(3))/40)1 \quad \text{(B.28)}$$

$$\begin{aligned}
 \tau &= \sqrt{504000 \sigma} \\
 &= \frac{504000 \sigma}{x(4) \times x(3)^2} \\
 &= \frac{tmpf}{x(4) \times x(3)^2}
 \end{aligned}
 \tag{B.31}$$

$$\begin{aligned}
 &= \frac{x(4)}{6} \\
 &= \frac{tmpf \times \sqrt{1 - \frac{1}{x(4)}}}{x(4)}
 \end{aligned}$$

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